^{34}S(n,n),(n,γ):resonances 1984Ca14

1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S. Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J, π, L, widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of Z=1-60.

^{35}S Levels

J^π, L, and resonance parameters E_n, $g\Gamma_n$, Γ_γ, and $g\Gamma_n\Gamma_\gamma/\Gamma$ are from 1984Ca14 due to significant discrepancies, including orders of magnitude inconsistencies, between the 1984Ca14 original data and 2018MuZY evaluation. $g\Gamma_n=(2J+1)\Gamma_n/2$.

E(level)[†]	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
6985.54?			0	−0.305	E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold. Γ_γ=0.38 eV.
7018.89 *4*		0.025 eV *1*	2	34.03 *1*	
7072.66 *5*		0.14 eV *4*		89.40 *3*	
7074.33 *6*		0.084 eV *3*		91.12 *4*	
7097.72 *30*		0.009 eV *3*	2	115.2 *3*	
7099.85 *20*		0.033 eV *5*	2	117.4 *2*	$g\Gamma_n$<1 eV.
7100.73 *4*	3/2$^-$			118.30 *1*	$g\Gamma_n$=372 eV *3*; Γ_γ=0.70 eV *4*.
7143.27 *11*		0.42 eV *1*		162.1 *1*	$g\Gamma_n$<1 eV.
7210.42 *5*			1,2	231.25 *2*	$g\Gamma_n$=62 eV *4*; Γ_γ=0.30 eV *2*.
7218.0 *4*		0.33 eV *2*		239.0 *4*	$g\Gamma_n$<4 eV.
7234.0 *5*		0.19 eV *1*		255.5 *5*	$g\Gamma_n$<4 eV.
7239.8 *6*		0.06 eV *2*		261.5 *6*	$g\Gamma_n$<2 eV.
7253.2 *6*		0.35 eV *2*		275.3 *6*	$g\Gamma_n$<1 eV.
7275.93 *5*	1/2$^+$			298.70 *3*	$g\Gamma_n$=9600 eV *150*; Γ_γ=2.82 eV *67*.
7279.13?		0.10 eV *3*		302	
7289.8 *8*		0.31 eV *3*		313.0 *8*	$g\Gamma_n$<6 eV.
7294.19 *6*	3/2$^-$			317.51 *5*	$g\Gamma_n$=2500 eV *25*; Γ_γ=0.23 eV *8*.
7331.00 *5*	1/2$^+$			355.41 *2*	$g\Gamma_n$=4810 eV *80*; Γ_γ=0.30 eV *30*.
7338.18 *20*			1,2	362.8 *2*	$g\Gamma_n$=230 eV *10*; Γ_γ=1.06 eV *7*.
7343.91 *20*		0.72 eV *3*		368.7 *2*	$g\Gamma_n$<2 eV.
7347.2 *5*		0.44 eV *6*		372.1 *5*	$g\Gamma_n$<2 eV.
7347.9 *5*		0.30 eV *5*		372.8 *5*	$g\Gamma_n$<2 eV.
7353.9 *5*		0.44 eV *4*		379.0 *5*	$g\Gamma_n$<10 eV.
7357.6 *5*		0.16 eV *4*		382.8 *5*	$g\Gamma_n$<6 eV.
7370.56 *6*	1/2$^-$			396.14 *4*	$g\Gamma_n$=6430 eV *150*; Γ_γ=3.08 eV *20*.
7371.97?		0.12 eV		397.6	
7396.0 *6*		0.44 eV *33*	2	422.3 *6*	
7404.6 *6*		0.21 eV *5*		431.2 *6*	$g\Gamma_n$<1 eV.
7408.68 *11*	3/2$^-$			435.4 *1*	$g\Gamma_n$=1570 eV *40*; Γ_γ=0.91 eV *8*.
7411.55 *6*				438.35 *5*	$g\Gamma_n$=49 eV *5*.
7416.50 *4*				443.45 *1*	$g\Gamma_n$=80 eV *8*; Γ_γ=0.51 eV *5*.
7429.1 *6*		0.12 eV *3*		456.4 *6*	$g\Gamma_n$<6 eV; Γ_γ=0.12 eV *3*.
7433.56 *5*				461.01 *2*	$g\Gamma_n$=50 eV *5*; Γ_γ=0.035 eV *20*.
7436.4 *4*			1,2	463.9 *4*	$g\Gamma_n$=296 eV *30*; Γ_γ=0.28 eV *3*.
7442.14 *5*	1/2$^+$			469.85 *2*	$g\Gamma_n$=1050 eV *35*; Γ_γ=0.37 eV *8*.
7462.2 *6*		0.33 eV *4*		490.5 *6*	$g\Gamma_n$=50 eV *10*.
7481.15 *6*	1/2$^-$			510.02 *4*	$g\Gamma_n$=375 eV *25*; Γ_γ=2.1 eV *1*.
7494.5 *6*		0.17 eV *4*		523.8 *6*	$g\Gamma_n$<4 eV.
7542.3 *6*		0.21 eV *3*		573.0 *6*	$g\Gamma_n$<1 eV.

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^{34}S(n,n),(n,γ):resonances 1984Ca14 (continued)

^{35}S Levels (continued)

E(level)[†]	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
7604.0 *6*		0.90 eV *8*		636.5 *6*	$g\Gamma_n$<10 eV.
7609.26 *5*	3/2$^-$			641.93 *3*	$g\Gamma_n$=2520 eV *60*; Γ_γ=0.51 eV *6*.
7648.9 *6*		0.21 eV *3*		682.7 *6*	$g\Gamma_n$<30 eV.
7655.7 *6*		0.80 eV *5*		689.7 *6*	$g\Gamma_n$<10 eV.
7663.91 *11*			1,2	698.2 *1*	$g\Gamma_n$=305 eV *50*; Γ_γ=0.80 eV *8*.
7678.3 *7*		0.45 eV *7*		713.0 *7*	$g\Gamma_n$<4 eV.
7731.0 *7*		0.31 eV *13*		767.3 *7*	$g\Gamma_n$<4 eV.
7749.7 *4*	(3/2)			786.5 *4*	$g\Gamma_n$=18 eV *6*; Γ_γ=2.65 eV *18*.
7761.46 *7*	3/2$^-$			798.65 *6*	$g\Gamma_n$=25350 eV *300*; Γ_γ=1.52 eV *50*.
7776.17 *11*	1/2$^-$			813.8 *1*	$g\Gamma_n$=860 eV *60*; Γ_γ=0.19 eV *13*.
7797.99 *9*	1/2$^+$			836.27 *8*	$g\Gamma_n$=3600 eV *160*; Γ_γ=0.39 eV *18*.
7812.24 *7*	3/2$^+$			850.94 *6*	$g\Gamma_n$=2650 eV *110*; Γ_γ=0.24 eV *19*.
7853.25 *7*	3/2$^-$			893.17 *6*	$g\Gamma_n$=4950 eV *160*; Γ_γ=0.38 eV *16*.
7861.8 *12*		1.13 eV *19*		902.0 *12*	$g\Gamma_n$<3 eV.
7880.8 *12*		0.67 eV *17*		921.5 *12*	$g\Gamma_n$<20 eV.
7889.0 *12*		0.57 eV *25*		930.0 *12*	$g\Gamma_n$<20 eV.
7894.63 *7*	3/2$^-$			935.78 *6*	$g\Gamma_n$=4100 eV *160*; Γ_γ=3.48 eV *23*.
7899.70 *30*				941.0 *3*	$g\Gamma_n$=176 eV *16*; Γ_γ=0.5 eV *3*.
7934.1 *15*		0.5 eV *3*		976.4 *15*	$g\Gamma_n$<2 eV.
7939.5 *15*		0.89 eV *4*		982.0 *15*	$g\Gamma_n$<8 eV.
7954.96 *11*	3/2$^-$			997.9 *1*	$g\Gamma_n$=930 eV *80*; Γ_γ=3.51 eV *19*.
7974.19 *11*	3/2$^-$			1017.7 *1*	$g\Gamma_n$=4020 eV *180*; Γ_γ=1.22 eV *17*.
8019.5 *2*			2	1064.4 *2*	$g\Gamma_n$=1032 eV *90*.
8041.20 *20*			2	1086.7 *2*	$g\Gamma_n$=2650 eV *160*.
8076.46 *20*			2	1123.0 *2*	$g\Gamma_n$=5480 eV *388*.
8077.91 *30*			2	1124.5 *3*	$g\Gamma_n$=4730 eV *320*.
8093.0 *5*	1/2$^-$			1140.0 *5*	$g\Gamma_n$=34880 eV *830*.
8143.66 *20*			2	1192.2 *2*	$g\Gamma_n$=5510 eV *290*.
8187.17 *30*	1/2$^-$			1237.0 *3*	$g\Gamma_n$=2945 eV *230*.
8211.25 *30*				1261.8 *3*	$g\Gamma_n$=975 eV *86*.
8224.2 *4*	1/2$^-$			1275.1 *4*	$g\Gamma_n$=2170 eV *180*.
8228.3 *4*			2	1279.4 *4*	$g\Gamma_n$=1140 eV *100*.
8243.98 *30*			2	1295.5 *3*	$g\Gamma_n$=1980 eV *150*.
8256.31 *20*	3/2$^-$			1308.2 *2*	$g\Gamma_n$=12475 eV *770*.
8262.4 *4*	5/2$^+$			1314.5 *4*	$g\Gamma_n$=6240 eV *500*.
8273.3 *4*	(1/2$^-$)			1325.7 *4*	$g\Gamma_n$=1635 eV *145*.
8298.27 *30*			2	1351.4 *3*	$g\Gamma_n$=14470 eV *650*.
8333.8 *4*	3/2$^-$			1388.0 *4*	$g\Gamma_n$=28845 eV *1057*.
8335.9 *4*			2	1390.1 *4*	$g\Gamma_n$=1955 eV *170*.
8391.3 *4*	3/2$^-$			1447.2 *4*	$g\Gamma_n$=27100 eV *1390*.
8393.64 *30*			2	1449.6 *3*	$g\Gamma_n$=3025 eV *260*.
8406.55 *30*			2	1462.9 *3*	$g\Gamma_n$=3885 eV *345*.
8418.8 *4*			2	1475.5 *4*	$g\Gamma_n$=3795 eV *340*.

† From $E_{c.m.}$+S(n)(^{35}S) where S(n)(^{35}S)=6985.84 *4* (2021Wa16) and $E_{c.m.}$=E_n(lab)×m(^{34}S)/[m_n+m(^{34}S)].